

MASTER BUILD PLAN v2

India-First AI Nutrition + Strength Training Platform

Nut AI foundation | Android + Web | Offline-first | Photo + Chat + Voice | Indian household-food intelligence | Workout journal | Equipment-aware training | Optional local/cloud AI | Cross-device sync

Core principle: AI interprets what the user means or what a photo contains. Verified data, recipes, portion logic, training records and deterministic code remain the source of truth.

This revision incorporates the full nutrition plan, workout journal, home-equipment model, custom exercises, progress analytics, architecture corrections, privacy/sync fixes, and the current decision to use IFCT during the private personal-use phase while keeping the data layer replaceable before public release.

Area	Current decision
Foundation	Fork and extend Nut AI; preserve its open-source license obligations.
Nutrition	India-first food ontology + IFCT for the current private build + USDA and other adapters + household recipes.
Training	Full set/rep/load journal, routines, programs, PRs, progression, charts and equipment-aware suggestions.
Home workouts	No separate section. Same exercise engine; equipment and location are attributes.
Offline	Core nutrition and training features work with zero API credits and no internet.
AI	Optional local multimodal AI and optional cloud AI. AI never owns final nutrition totals or workout history.
Platforms	Expo/React Native mobile app + synchronized desktop/web app.

1. Product Vision

Build one personal performance platform that connects what the user eats, how bodyweight changes, and how training performance progresses. The experience should be quick enough to use during a workout or meal, but deep enough to support serious analysis later on the web.

- Nutrition: photo, gallery, web upload, chat, voice, barcode, label, manual search and saved meals.
- Training: set-by-set logging, previous-performance context, rest timing, progression, PRs, exercise history, routines/programs and analytics.
- India-first: Indian food names, regional dishes, household measures, oil/ghee uncertainty and household recipe learning.
- Equipment-aware: the app understands what equipment the user owns rather than separating gym and home training.
- Offline-first: losing internet or API balance does not break the tracker.
- Unified history: food, weight, body measurements and training belong to one daily timeline.

North-star: a nutrition and training app that learns how your home cooks and how you train.

2. Navigation and UX Structure

Tab	Purpose
Home	Today dashboard: calorie/protein status, weight trend, today's workout, recent PRs and active workout mini-card.
Food	Meals, camera/gallery, chat/voice, recipes, barcode, manual foods.
Train	Active workout, routines, programs, exercise library, equipment-aware suggestions.
Progress	Strength, body, nutrition and combined long-term analytics.
You	Profile, goals, sync, AI, equipment, custom exercises, utensils, privacy, export/import.

An active workout survives navigation and app backgrounding. If the user opens Food mid-session, show a persistent mini-card with workout duration/rest timer and one-tap return to Train.

3. Unified Nutrition Input Pipeline

Camera / Gallery / Web Upload / Chat / Voice / Barcode / Label / Manual | v Structured observations | +-----+-----+ | Known information Unknowns | | clarification engine +-----+-----+ v resolver -> recipe engine -> gram engine | deterministic nutrition | confidence + provenance | meal log

Every input route creates the same structured meal representation. Photo and chat are interfaces, not separate calorie systems.

4. Zero-API / Offline Nutrition Mode

- Local food search and aliases.
- Manual meal creation/editing.
- Calories/macros from stored data.
- Custom foods, recipes, household recipes and saved meals.
- Indian units: katori, spoon, glass, piece, count.
- Weight, goals, BMR/TDEE, progress history.
- Barcode/local product lookup where data exists.
- Deterministic parser for simple text such as "2 roti 100g rice 1 katori dal".
- JSON export/import and local backup.
- Structured corrections and recipe calculations without AI.

Without AI, difficult free-form language and photo understanding can be unavailable or reduced, but the actual tracker remains functional.

5. AI Architecture

5.1 Local AI

Current Android candidate: Gemma 3n E2B as an optional downloadable local-AI pack. E4B can later be offered on devices that benchmark well enough.

- Use local AI for food-image component identification, difficult language parsing, correction interpretation and later optional voice understanding.
- Bridge from Expo/React Native through a native Android module/runtime; do not make Expo Go a production dependency.
- Do not bundle a multi-GB model in the base app. Download/remove on demand.
- Benchmark latency, RAM, thermals and Indian-food recognition before enabling by default.

5.2 Cloud AI

Use a provider adapter and benchmark cheap multimodal models on the same Indian-food test set. Initial development candidates remain Gemini Flash-class and a low-cost OpenAI multimodal model. Keep API usage sparse: deterministic parser first, then local AI, then cloud AI if needed and available.

5.3 AI confidence correction

Do not treat a model-generated numeric confidence as calibrated truth. Final confidence should combine recognition ambiguity, nutrition-source quality, portion evidence, recipe uncertainty, hidden-ingredient uncertainty and empirical evaluation error.

6. India-First Nutrition Source Layer

All nutrition sources sit behind a common adapter so no single database becomes architectural glue.

NutritionSource searchFoods() getFood() getNutrients() getSourceMetadata() getLicenseInfo() Providers: IFCTSource USDASource OpenFoodFactsSource UserFoodSource HouseholdRecipeSource

6.1 Current personal-use decision for IFCT

For the current private build intended for personal use, use IFCT as a primary Indian nutrition source. Keep it behind IFCTSource so public/commercial distribution can later be reviewed, permitted, replaced or selectively disabled without rewriting the application.

Release checkpoint: before public or commercial distribution, re-check NIN permission/redistribution requirements and the exact way IFCT data would be bundled or exposed. This is an implementation guardrail, not a blocker for the current private prototype.

6.2 Source separation

- Keep USDA, IFCT, Open Food Facts, user foods and household data logically separated with explicit provenance.
- Do not collapse them into an opaque master table with lost licensing/source metadata.
- Resolver may search all sources while each record retains source, retrieval/import version and attribution metadata.

7. Indian Food Ontology and Recipe Engine

- Canonical IDs with English + Hinglish + Hindi aliases first; regional languages later.
- Map arhar/toor/tuvar/pigeon pea; roti/chapati/phulka; dahi/curd; preserve preparation differences.
- Preserve regional identities such as litti, chokha, sattu dishes, poha variants, dosa variants and biryani variants.
- Separate dish family from specific household recipe and actual meal instance.

Serving nutrition = total recipe nutrition x serving weight / final cooked recipe weight.

Store ingredient quantities, cooking method, added fat, water/yield changes, final cooked weight and serving size. Generic priors fill missing information; household-confirmed data progressively replaces generic priors.

8. Hidden-Ingredient and Clarification Engine

Question value = expected reduction in nutrition uncertainty / user effort.

Unknown	Typical impact	Behavior
Oil/ghee in curry	Very high	Ask early.
Serving/portion	High	Ask or use utensil/geometry.
Cream/cashew/coconut	High in relevant dishes	Ask conditionally.

Unknown	Typical impact	Behavior
Dal type	Often smaller	Ask only if materially useful.
Rice variety	Often small	Usually infer/default.
Minor spices	Tiny	Do not interrupt.

Always offer "I don't know". Use a prior and widen the range rather than forcing fake information.

9. Household Learning and Measures

- Learn typical roti size/atta weight and normal ghee usage.
- Learn rice grams for the user's usual katori.
- Learn household dal type/concentration/tadka/batch servings.
- Learn recurring sabzi/curry recipes and packaged products.
- Store provenance: measured, confirmed, learned, inferred, generic or unknown.
- A one-off correction applies to that meal first. Only promote to a household default after explicit confirmation or repeated consistent evidence.
- Version household recipes rather than rewriting history.

9.1 Utensils and density

- Support katori, bowl, glass, cup, tsp, tbsp, ladle, serving spoon, handful, piece/count.
- Allow personal utensils with name, volume/size and optional reference photo.
- Build India-specific density/portion references for cooked rice, thin/thick dal, rajma, chole, poha, upma, curd, sambar and gravies.

10. Portion Estimation

- Exact packaged serving.
- Discrete count.
- Learned household portion.
- Known utensil/plate geometry.
- Image/reference geometry.
- Volume x food density.
- Standard Indian portion prior.
- AI/model estimate as weaker evidence.

Strongly conflicting estimates should not be silently averaged. Preserve Nut AI's trust-hierarchy philosophy: choose stronger evidence, ask the user or surface the disagreement. Support multiple photos per meal for layout, depth and close-up recognition.

11. Training System - Core Philosophy

Training is not a bolted-on "workout calories" feature. It is a full strength-training journal sharing the same user, timeline, sync and analytics foundation as nutrition.

During training: optimize for one-thumb speed. After training: provide deep history and graphs.

- Previous workout values shown directly beside current set entry.
- Auto-fill/duplicate sets to minimize typing.
- Automatic rest timer after set completion.
- Planned sets and actual sets stored separately.
- Workout session persists locally even if app backgrounds or process restarts.
- All training remains usable offline.

12. Exercise and Set Data Model

12.1 Exercise tracking types

Tracking type	Examples
Weight + reps	Bench press, squat, curl.
Bodyweight + reps	Push-up, pull-up.
Assisted bodyweight	Assisted pull-up/dip.
Reps only	Some calisthenics/drills.
Duration	Plank, holds.
Distance + duration	Run, walk, cycle.
Weight + duration	Loaded carry/hold.
Custom	User-defined tracking pattern where supported.

12.2 Set types

- Warm-up
- Working
- AMRAP
- Failure
- Drop set
- Rest-pause
- Assisted
- Optional tempo prescription

Advanced set types should stay out of the default visual path; change via set-menu/long-press. Warm-ups are excluded from normal hard-set/working-volume analytics.

12.3 Planned vs actual

Store plannedWeight/plannedReps/plannedRIR separately from actualWeight/actualReps/actualRIR. This preserves program intent and enables adherence analytics.

13. Custom Exercises - No "Every Exercise on Earth" Problem

Ship a useful built-in exercise library, but allow custom exercises to behave exactly like built-in ones in routines, history, PRs, progression and charts.

- If search misses an exercise, show + Create "searched name".
- Custom exercise fields: name, tracking type, primary muscles, secondary muscles, equipment, optional notes/media.
- Custom exercises are saved for future use and sync like normal entities.
- Exercise aliases help avoid duplicates.

- Never automatically merge PR history for genuinely different variants such as free-weight bench vs Smith bench.

14. Home Workouts and Equipment Model

There is no separate Home Workouts section. A curl at home and a curl at a gym use the same exercise engine. Location is optional metadata; equipment availability is what matters.

14.1 Equipment inventory

Support a personal equipment profile such as: dumbbell handles x2, EZ bar x1, short straight bar x1, 5 kg plates x6, 2.5 kg plates x4. Users can add arbitrary custom equipment.

- Equipment inventory filters exercise suggestions.
- Programs can be generated/filtered for "available today" equipment.
- Workout location may be Home/Gym/Park/Hotel/Other without creating separate analytics silos.
- Adjustable dumbbell loads store whether weight is per-hand or total external load.

14.2 Personalized plate calculator

Given bar weight, target weight and the user's actual plate inventory, calculate a symmetric loadout. If the requested target cannot be assembled, show nearest valid weights instead of suggesting impossible loading.

15. Active Workout UI/UX

BARBELL BENCH PRESS Last time: 60x10 | 60x9 | 57.5x10 Set Previous kg reps RIR W 20x10 20 10 - [done] 1 60x10 62.5 9 2 [done] 2 60x9 62.5 __ [] 3 57.5x10 60 __ [] + Add Set Rest 01:37 History | Graph | Replace | Note

- Large numeric inputs and one-thumb controls.
- Duplicate/pre-fill sets.
- Rest timer auto-start on completion; +15/-15/skip.
- Exercise-specific rest defaults.
- Permanent exercise note, workout-specific exercise note and individual set note.
- Superset/circuit visual grouping.
- Replace exercise by equipment/muscle similarity without merging histories.

16. RIR/RPE and Effort

RIR (reps in reserve) and RPE are optional advanced inputs, not mandatory beginner clutter. Store set RIR when useful and ask one simple session-RPE (1-10) after a workout if the user enables it.

The app should treat RIR/RPE as subjective effort data, not laboratory truth. They complement load/reps and help explain why identical external volume can feel very different.

17. PR Engine and Exercise History

PR/metric	Examples
Heaviest load	100 kg.
Estimated 1RM	Calculated from appropriate submaximal sets and clearly labeled estimated.
Rep-range PRs	Best 3, 5, 8, 10 rep performance.
Best set volume	Load x reps for one set.
Highest session volume	Exercise-specific session total.
Bodyweight exercise PR	Most reps / weighted variant.
Duration PR	Longest plank/hold.
Cardio PR	Distance, pace or time depending on activity.

Do not use very high-rep sets as if they precisely predict 1RM. Keep the formula configurable internally and show estimation limitations.

18. Progression Engine

Progression suggestions are recommendations, never silent program edits.

- Double progression (e.g. reach all sets at top of rep range, then increase load).
- Fixed load increment.
- Percentage progression.
- RIR-based progression.
- Manual progression.
- Program-defined rules.

Example: 3 sets of 8-12 at target RIR 2. Once the user reaches 12/12/12 within target effort, suggest the next achievable load based on equipment availability.

19. Training Analytics - Do Not Worship One Metric

Track volume-load (sets x reps x load), but do not present it as the master progress score. Strength, hard sets, frequency, effort and exercise-specific performance answer different questions.

Dimension	Metrics
Strength	e1RM, top-set load, rep PRs.
Hypertrophy-oriented workload	Hard working sets per muscle/week, frequency, RIR distribution.
External workload	Sets, reps, volume-load.
Internal effort	Set RIR/RPE, session RPE.
Consistency	Sessions/week, planned vs completed.

19.1 Muscle-group volume

Show direct sets and optionally fractional/effective sets for secondary muscles. Make the approximation explicit; do not pretend fractional set weighting is a biological measurement.

20. Recovery and Readiness

Avoid fake precision such as "Chest 73.2519% recovered." Start with transparent context: last trained, hard sets in last 7 days, session RPE, soreness and optional readiness inputs.

- Energy 1-5
- Soreness 1-5
- Sleep quality
- Motivation/readiness

Later, if an estimate is added, use labels such as Low / Moderate / High recovery/readiness with explanation rather than spurious decimal certainty.

21. Workout Tools

- Automatic rest timer with notifications/vibration.
- Warm-up calculator with customizable templates; warm-ups excluded from working volume.
- Plate calculator based on actual owned plates/bar weights.
- Superset/circuit support.
- Exercise substitution by muscle/equipment.
- Calendar view for training history and program schedule.

22. Routines, Programs and Quick Workouts

Object	Meaning
Quick workout	Start immediately and add anything.
Routine	Reusable workout template such as Push/Pull/Legs.
Program	Multi-week sequence with planned progression and adherence.

Programs store intent; workouts store what actually happened. The web app should provide a richer drag/drop program builder while mobile remains optimized for execution.

23. Combined Progress Dashboard

Area	Examples
Overview	Bodyweight trend, key lifts, protein average, training adherence.
Strength	e1RM/load/reps/volume/frequency graphs and PRs.
Body	Weight, waist, chest, arms, thighs, optional body-fat estimate, progress photos.
Nutrition	Calories, protein, carbs, fat, bodyweight trend and goal adherence.
Combined	Long-term nutrition/bodyweight/training trends with careful non-causal wording.

Do not claim that a nutrition change caused a strength change merely because the graphs moved together. Present associations and let the user inspect context.

24. Exercise Calories and Nutrition Integration

Do not automatically eat back workout-calorie estimates. Resistance-exercise energy expenditure and wearable estimates can be noisy. Show exercise energy as informational if available, but keep adaptive nutrition driven primarily by intake + bodyweight trend over time.

Optional training-day/rest-day targets are fine when the user explicitly chooses them, such as stable protein with intentional carbohydrate/calorie differences.

25. Training Chatbot

One assistant spans food and training, but it queries structured records rather than remembering history from chat messages.

- "What did I bench last push day?"
- "Show my deadlift progress this month."
- "How many chest sets this week?"
- "Replace today's pull workout using only my home equipment."
- "Compare my protein average with my training weeks."

Chat produces queries or structured operations; database state remains authoritative.

26. Mobile + Web Monorepo

apps/ mobile/ web/ packages/ core-schema/ perception/ conversation/ indian-ontology/ indian-data/ recipe-engine/ resolver/ gram-engine/ clarification/ household-memory/ totals/ goals/ provenance/ training-core/ exercise-library/ workout-session/ progression/ training-analytics/ training-plans/ health-connect/ sync-schema/ eval/

Keep shared domain packages pure TypeScript wherever practical. Native React Native/Android dependencies stay in apps/mobile or platform adapters.

27. Mobile and Web Responsibilities

Mobile

- Camera + gallery, including already-clicked photos.
- Barcode/label, fast chat/voice.
- Active workouts, rest timer, set logging, equipment-aware plate calculator.
- SQLite immediate source of local truth.
- Optional downloadable local AI.
- Offline queue and background-safe session persistence.

Web/Desktop

- Drag/drop/file-picker image upload and later clipboard paste.
- Full meal/recipe editor.

- Program builder and exercise library management.
- Large training/nutrition/body analytics.
- Same chat/correction system.
- Cross-device synchronized account.

28. Sync, Local-Only Mode and Migration

Cloud sync remains optional. Mobile writes locally first; network never blocks normal logging. A

Postgres/auth/object-storage backend such as Supabase is a practical first implementation candidate, but domain code should remain provider-independent.

- Stable UUIDs, updatedAt, revision, deletedAt and syncState.
- Idempotent offline queue.
- Structured operation log only where history matters: meal corrections, recipe edits, workout edits.
- Store photos in object storage, not database blobs.
- Design local -> account migration: claim the anonymous local dataset, upload, deduplicate and assign account ownership.
- Define simple deterministic conflict rules first; do not over-engineer CRDTs before real usage requires them.

29. Photo and Body-Data Privacy

- Before cloud analysis/storage: resize/compress as appropriate and strip EXIF/GPS metadata.
- Let user choose Keep meal photos or Delete after analysis.
- Progress photos local/private by default; cloud upload only with explicit sync choice.
- Do not embed shared production API keys in APK/web bundles. Hosted AI uses server-side credentials; BYO keys use secure storage.

30. Health Connect Strategy

Our own workout database remains authoritative. Health Connect is an adapter for importing/exporting supported exercise/session data. Capability-gate newer fields such as set index/weight/RPE instead of designing the whole schema around whatever a specific Health Connect version currently exposes.

31. Core Data Model

Entity	Responsibility
Food	Canonical identity, aliases, nutrients, preparation, source.
Dish Family	Regional identity and generic priors.
Recipe / Household Recipe	Ingredients, method, yield, source/version and personal overrides.
Meal / Meal Component	Time/type, components, amounts, totals, uncertainty and provenance.
Exercise / Custom Exercise	Tracking type, muscles, equipment and identity.
Equipment / Plate Inventory	Owned/custom equipment and loadability.
Routine / Program / Program Day	Reusable and multi-week training plans.
Workout Session / Exercise / Set	Planned vs actual training data, rest, RIR/RPE and notes.
PR / Training Metric	Exercise-specific records and analytics.
Body Measurement	Weight, measurements and private photo references.
Operation	Structured edit for sync/audit/undo.

32. Evaluation Programs

32.1 Indian food golden dataset

- Weighed ingredients, added fats, final cooked yield and served portion.

- Top/angled/side photos across devices/lighting.
- Simple foods, thalis, curries, fried foods, breads, regional dishes, restaurants and packaged food.
- Food identification top-1/top-3; portion gram MAE/MAPE; calorie/macro MAE/MAPE.
- Calibrate uncertainty from observed error rather than model self-confidence.

32.2 Training analytics tests

- Unit tests for e1RM formulas, PR detection, rep-range records and volume calculations.
- Test warm-up exclusion and set-type behavior.
- Test per-hand vs total dumbbell load conventions.
- Test plate calculator against inventory and symmetry.
- Test program adherence and planned/actual separation.
- Test exercise replacements never corrupt historical PR identity.

33. Accuracy / Effort Modes for Food

Mode	Experience
Quick	One photo/chat entry, minimal interruption, wider honest range.
Accurate	1-3 highest-value questions plus household/geometry evidence.
Precise	Measured recipe/weights/portions; narrowest uncertainty.

34. Open-Source and Licensing Strategy

- Use Nut AI under its applicable AGPL obligations and preserve notices/source availability.
- Current IFCT use is for the private personal build; keep IFCT behind an adapter and perform a release-permission checkpoint before public/commercial distribution.
- Track every code/data source independently in LICENSES/NOTICE metadata.
- Keep Open Food Facts provenance/license behavior separate from other stores.
- Do not assume a public GitHub dataset automatically grants rights to underlying third-party data.
- If the project later becomes commercial, review distribution terms before release rather than blocking personal development now.

35. Revised Build Roadmap

#	Phase	Deliverable
0	Fork/reproduce Nut AI	Build Android app, run tests, inventory code/data/licenses.
1	Offline core hardening	Manual nutrition tracker, history, goals, weight, backup work with zero API.
2	Shared schemas	Stabilize food/meal/recipe/training/provenance/sync contracts.
3	IFCT + Indian data layer	Private-build IFCTSource, aliases, units, USDA fallback and resolver scoring.
4	Indian recipe/portion engine	Dish families, yield, hidden fat, katori/density and household recipes.
5	Workout journal MVP	Exercise library, custom exercises, set logging, previous values, rest timer, offline persistence.
6	Equipment system	Inventory, custom equipment, plate calculator and equipment-aware exercise filtering.
7	Training analytics	PRs, e1RM, exercise graphs, muscle sets, session RPE and progression suggestions.
8	Clarification + offline text	Nutrition uncertainty questions and deterministic Hinglish-style parser.
9	Cloud photo/chat AI	Structured multimodal perception, corrections, strict schemas and provider adapter.
10	Sync foundation	Auth, Postgres, storage, offline queue, local-account migration and conflicts.
11	Web app	Food logging, uploads, workout/program editor, analytics and sync.
12	Household learning	Promote repeated confirmed recipes/portions with provenance and versioning.
13	Food golden dataset	Weighed Indian evaluation meals and automated metrics.

#	Phase	Deliverable
14	Local AI prototype	Gemma 3n E2B Android integration after product core is proven.
15	Health Connect	Capability-gated import/export adapter.
16	Beta hardening	Migrations, security, privacy, performance, recovery and test matrix.
17	Public-release checkpoint	Re-check IFCT/public data permissions, notices and redistribution strategy.
18	Open-source release	Docs, contributor setup, privacy docs and reproducible builds.
19	Commercial phase later	Hosted AI/sync/premium services while preserving applicable licenses.

36. Revised MVP

- Android app based on Nut AI foundation.
- Fully usable offline/no-AI nutrition tracker.
- Private-build IFCT Indian food layer plus USDA fallback.
- Indian recipe, oil/ghee, katori and household recipe support.
- Camera + gallery food input and cheap cloud multimodal recognition.
- Hinglish chat logging/corrections and high-value clarification.
- Workout journal: built-in + custom exercises, sets/reps/load, previous performance, rest timer and notes.
- Equipment inventory and personalized plate calculator.
- Exercise history, core PRs and progress graphs.
- Household recipe/portion saving.
- Backup/export/import.

MVP proof: the user can log a normal Indian home meal and a full strength workout offline, use AI when available for convenience, correct both domains without losing provenance, and see meaningful progress over time.

37. Post-MVP Priorities

- Full phone/web sync and desktop program builder.
- Local Gemma 3n E2B mode.
- Voice logging.
- More regional food ontology and multilingual aliases.
- Multi-view/depth-assisted portion estimation.
- Automatic household learning.
- Advanced progression/program rules.
- Health Connect integration.
- Unified assistant over structured nutrition/training history.
- Community contributions only with provenance/moderation/license controls.

38. Engineering Guardrails

- AI never owns final nutrition totals or historical workout facts when deterministic records exist.
- Every nutrition number carries provenance; every training metric has a defined formula.
- Strongly conflicting portion evidence is surfaced/resolved, not quietly averaged.
- Every AI/cloud path has timeout/failure/fallback behavior.
- Workout session state survives navigation/background/process restart.
- Database migrations and backups are tested.
- Shared schemas are runtime validated.
- No impossible plate/load suggestions: progression consults equipment inventory.

- Warm-ups do not inflate working-set analytics.
- Exercise calories do not automatically raise food budget.
- Private photos remain private by default.
- Build the deterministic core before polishing advanced AI.

39. First Development Sprint

Order	Task	Done when
1	Fork/run Nut AI	Android development build launches; test status documented.
2	Architecture + license map	Reusable/replaced packages and source adapters documented.
3	No-AI regression	Meal log/edit/search/save/export works with no provider key.
4	Shared schemas	Food/recipe/meal/exercise/workout/provenance schemas validate at runtime.
5	IFCT private adapter	Initial IFCT records import/search/resolve behind NutritionSource.
6	Indian aliases	100+ high-frequency aliases resolve deterministically.
7	Recipe prototype	Roti, rice, dal, sabzi and one rich curry calculate from recipe/yield.
8	Workout logger	Create workout, custom exercise, sets/reps/load, finish and reopen history offline.
9	Equipment profile	Enter bars/plates and calculate valid loadouts.
10	Progress basics	PR detection and one exercise progress graph work.

40. Final Product Definition

The completed product accepts food through camera, existing photos, web uploads, chat, voice, barcode, label, manual search or saved recipes; and accepts training through quick workouts, routines and programs. Both domains share the same user, timeline, offline-first storage, sync and analytics system.

The nutrition engine resolves trustworthy data, models recipes/portions, asks only useful questions and learns household cooking. The training engine records real set-level performance, understands equipment, supports custom exercises, suggests progression without overriding the user, and exposes multiple dimensions of progress rather than one misleading score.

If cloud AI disappears or API credits reach zero, the application remains a functional nutrition and training journal. If the user trains at home with a pile of plates and an EZ bar instead of a gym membership, the same system still works. That is the architecture we want.

41. Agreed Decision Checklist

Decision	Status
Nut AI open-source foundation	Agreed
Private-build IFCT use	Agreed
Public-release IFCT permission checkpoint	Agreed
USDA/other nutrition adapters remain modular	Agreed
Zero-API usable tracker	Required
Camera + gallery + web uploads	Required
Hinglish chat/corrections	Required
Indian recipes + oil/ghee logic	Required
Household learning	Required
Workout journal with sets/reps/load	Required
No separate Home Workout section	Agreed
Custom exercises/equipment	Required
Personal plate inventory/calculator	Required
RIR/RPE optional	Agreed

Decision	Status
PRs/progression/graphs	Required
Exercise calories not auto-eaten-back	Agreed
Web + phone synchronization	Required
Local-only privacy mode	Required
Local AI later, not blocking MVP	Agreed
Measured Indian food evaluation dataset	Required before serious accuracy claims

42. Research Notes / Source Pointers

These are implementation references used during planning. Re-check current terms/API versions before coding or release.

- Nut AI repository: <https://github.com/Blueturboguy07/nut-ai>
- ICMR-NIN / IFCT: https://www.nin.res.in/ebooks/IFCT2017_16122024.pdf
- USDA FoodData Central: <https://fdc.nal.usda.gov/>
- Open Food Facts licensing/API: <https://openfoodfacts.github.io/openfoodfacts-server/api/>
- Android Health Connect workouts:
<https://developer.android.com/health-and-fitness/health-connect/experiences/workouts>
- Hevy feature/PR references: <https://help.hevyapp.com/>
- Fitbod recovery reference: <https://help.fitbod.me/>
- ACSM resistance training position stand / PubMed: <https://pubmed.ncbi.nlm.nih.gov/41843416/>
- RIR validity review / PubMed: <https://pubmed.ncbi.nlm.nih.gov/36135029/>

MASTER BUILD PLAN - V3 ADDITIONS

Agreed product foundations added after the v2 plan

Status: These items are now part of the agreed product plan. They are not optional brainstorming notes.

23. Data Completeness and Adaptive Nutrition

Adaptive calorie recommendations must never treat an incomplete food log as a fully observed day. Otherwise, the expenditure model learns from missing dinners and becomes confidently wrong.

- Every day stores a nutrition logging status: COMPLETE, PARTIAL, UNKNOWN, or FASTING.
- Only complete or explicitly trusted days should materially influence adaptive energy-expenditure calculations.
- Partial days are excluded or down-weighted rather than interpreted as low-calorie days.
- The UI should make this low-friction, for example a single end-of-day confirmation: 'Finished logging today?'
- Adaptive recommendations should use intake history + bodyweight trend + time, not smartwatch/workout calorie estimates.

23.1 Weekly check-in

Add an optional weekly check-in that summarizes weight trend, average intake, protein, training adherence and strength movement. The app may recommend a small calorie adjustment, but the user must explicitly accept it.

Weekly check-in - Weight trend - Average calories - Estimated expenditure - Protein average - Planned vs completed training - Strength trend - Suggested calorie change - Keep / Accept

24. Fast Repeat Logging

Repeated logging is a core UX requirement. AI should not force the user to re-analyze food or re-type training they repeat constantly.

- Food: Recent, Frequent, Saved Meals, My Recipes, Yesterday, Favorites.
- One-tap actions such as 'Copy yesterday's breakfast' and 'Log usual lunch'.
- Training: prefill previous set values beside today's empty set.
- Duplicate set / duplicate multiple sets.
- Recently used exercises and foods rank higher in search.
- All repeat actions operate offline.

25. Universal Search

The app needs a fast fuzzy search layer that understands aliases, recent use and user-created data.

Search context	Behavior
Food	Find canonical foods, aliases, recipes, household foods, recent logs and custom foods.
Training	Find built-in exercises, aliases and custom exercises.
Language	Support common English/Hinglish/Hindi variants and spelling errors.
Ranking	Prefer exact matches, user-created entries, favorites and recent/frequent use.
Offline	Maintain a local searchable index; cloud search is not required for core use.

Examples such as arhar/arahar/toor/tuvar should resolve naturally without requiring the user to know the canonical database name.

26. Unified Daily Timeline

Food, weight and training should appear in one chronological daily history instead of separate data islands.

TODAY 07:42 Weight 59.2 kg 08:16 Breakfast - 612 kcal / 34 g protein 12:58 Lunch - 744 kcal / 29 g protein 16:21 Pre-workout
17:08 Push A started 18:22 Push A completed - PR 20:04 Dinner

- Timeline items link directly to their underlying meal, workout or measurement.
- Supports edits/undo from history.
- Web dashboard and mobile Today screen use the same event model.
- Future Health Connect imports can appear in the same timeline.

27. Weekly and Monthly Reports

Reports should summarize real measurements rather than generate a vague 'health score'.

- Average bodyweight and trend.
- Average calories and protein.
- Number of complete nutrition-log days.
- Training sessions planned vs completed.
- Working sets and frequency.
- PRs and best exercise improvements.
- Strength trends such as estimated 1RM.
- Consistency/adherence.
- Optional notes on large changes or missing data.

Reports should work locally from structured history; AI may later explain them, but the numbers themselves remain deterministic.

28. Progressive Disclosure: Simple vs Advanced Training UX

The training engine may be sophisticated, but beginners should not be forced to understand every training metric.

Mode	Default visible fields
Simple	Exercise, weight, sets, reps, previous performance, rest timer.
Advanced	RIR/RPE, set type, progression rules, tempo, deeper analytics and programming fields.

Advanced fields are optional and can be enabled progressively. The design principle is: complex engine, simple surface.

29. Micronutrient-Ready Nutrition Model

Even if V1 primarily shows calories/macros, the database schema should support micronutrients from day one to avoid a destructive migration later.

- Support vitamins, minerals, fiber, sodium, potassium and other nutrients when the source provides them.
- Every nutrient value retains its source/provenance.
- Missing nutrient values must remain unknown rather than silently becoming zero.
- Show nutrient-data coverage/completeness when displaying detailed micronutrient totals.
- Allow future nutrient targets such as floor, target or ceiling.

Example: if only 72% of the day's foods contain reliable B12 values, show that coverage rather than pretending the reported B12 total is complete.

30. Health Connect Integration

Use Android Health Connect as the external bridge for health data rather than rebuilding commodity trackers.

- Potential imports: weight, steps, sleep, exercise sessions and selected fitness metrics.
- Potential exports: workouts where the platform/device supports the required record fields.
- Our database remains authoritative for detailed resistance-training data.
- Health Connect capabilities are adapter-based and capability-gated; do not shape the core schema around unstable platform fields.
- Imported data stores source metadata to prevent duplicate records.

31. Undo, Versioning and Safe Editing

All destructive or high-value edits need a recovery path.

- Delete meal/workout -> immediate Undo.
- Meal quantity/correction -> Undo.
- Recipe edits -> version history.
- Household recipe changes -> new version, never silently rewrite historical calculations.
- AI corrections produce structured operations that can be reverted.
- Soft-delete important synced records before final cleanup.

32. Data Quality Inspector

Power users should be able to tap 'Why this number?' and inspect how the result was produced.

Arhar dal Recipe: Home Dal v4 Portion: 168 g Oil: 2 tsp / 4 servings Nutrition source: IFCT (private build) Portion evidence: household katori Confidence: high Uncertainty contributors: small

- Show nutrition source.
- Show recipe version.
- Show portion evidence.
- Show hidden ingredient assumptions.
- Show measured/confirmed/inferred/generic status.
- Show uncertainty contributors rather than a meaningless model confidence percentage.

33. Active Workout Outside the Train Screen

An active workout must survive navigation and app backgrounding.

- Persistent mini workout card while the user visits Food, Home or Progress.
- Rest timer continues when navigating away.
- Android notification/live control for current exercise/rest timer in a later phase.
- Persist active session state to local storage/database, not just React component state.
- Restore an unfinished session after process death/crash.

Later live controls may expose set completion and +/- rest-time actions, but only after the base workout flow is reliable.

34. Updated Feature Priorities

Priority	Addition	Reason
P0	Day completeness	Protect adaptive calorie calculations from bad data.
P0	Fast repeat logging	High-frequency daily UX improvement.
P0	Weekly adaptive check-in	Connect intake to actual weight trend.
P0	Universal search	Required as foods/exercises/custom data grow.
P0	Undo/versioning	Protect valuable historical data.
P1	Unified daily timeline	Makes nutrition + training one coherent product.
P1	Weekly/monthly reports	Turns logs into useful feedback.
P1	Simple/advanced training UX	Keep beginners from drowning in metrics.
P1	Micronutrient-ready schema	Avoid painful future data migration.
P1	Health Connect	Import commodity health data instead of rebuilding it.
P2	Live workout controls	Excellent convenience after core stability.
P2	Data-quality inspector	Transparency and debugging/power-user value.

35. Explicit Non-Goals for the Current Product

- No social feed, follower counts or public leaderboard in the core roadmap.
- No generic challenge/gamification system unless it proves useful later.
- No supplement marketplace or shopping layer.
- No meal-delivery/grocery commerce.
- No exercise-form camera scoring in the current plan.
- No attempt to become a smartwatch OS, GPS running platform or sleep-analysis product.
- No meaningless single 'health score'.

The product remains focused on four jobs: understand what I eat, understand how I train, learn my habits, and show whether I am progressing.

36. Updated Personal-Build IFCT Decision

For the current private personal-use build, IFCT is approved as a primary Indian nutrition source in the implementation plan. It should still sit behind the modular NutritionSource interface.

- Current private build: IFCT may be used directly as the primary Indian source.
- USDA remains a complementary/fallback source.
- Open Food Facts stays provenance-separated because of its separate database license.
- Household recipes and user-measured foods remain higher priority when they are more specific.

- Before public/open-source redistribution or commercial launch, perform a dedicated IFCT permission/licensing checkpoint.
- If permission terms are unsuitable later, the adapter architecture allows IFCT to be disabled/replaced without rewriting the app.

37. Updated Build Order

These additions also change implementation order. Reliability and repeated-use UX come before local-AI spectacle.

Order	Workstream
1	Stable offline nutrition + training data model, migrations, undo and backups.
2	Indian food/recipe/portion engine with IFCT for the current private build.
3	Workout journal, custom exercises/equipment, routines, PRs and progression.
4	Fast repeat logging, universal search and unified daily timeline.
5	Day completeness + adaptive weekly nutrition check-in.
6	Cloud photo/chat AI and structured corrections.
7	Sync/account migration + web app.
8	Reports, micronutrient detail and Health Connect.
9	Measured Indian-food evaluation dataset and confidence calibration.
10	Local multimodal AI and live workout controls after the core is proven.

38. V3 Product Standard

The app should be judged by whether a person can use it daily for months without excessive typing, unexplained numbers, broken offline behavior or data corruption. New features should only enter the roadmap when they improve logging speed, accuracy, progression visibility or trustworthy decision-making.

Final principle: sophistication belongs in the engine; daily interaction should stay fast and calm.